DSAI4901 · CAPSTONE PROJECT PROPOSAL

# Paper-Distillation PPT Agent

## From generating to verifying

A slide generation system built around content traceability and deck-readiness

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**38**
papers reviewed

**8**
functional gaps

**1**
research direction

# Literature review: from 367 papers to 38

**367** arXiv candidates

**69** after relevance screening

**38** read in depth (33 agents + 5 education)

**18** with author-released code

18/38 release author code (each link verified by HTTP); the rest are paper-only or anonymous submissions

### WHAT I ACTUALLY DID

- 12 arXiv queries across slide, PPT, poster and deck
- Relevance screening to remove unrelated work
- Full-text keyword scan of all 35 papers
- GitHub links extracted from each PDF, then verified by HTTP

# Four things they already do well

First establish what is solved — this decides what I should not rebuild

01

## Generation & layout

Decks already look convincing: stable layout, consistent styling, competitive in blind tests.

Zheng et al. (2025); Vats et al. (2026)

02

## Native editable output

Work has moved into the .pptx object layer, preserving styling and editability.

Jung et al. (2025); Zheng et al. (2026b)

03

## Evaluation methods

From holistic scores to task-based tests: answering questions on a deck, gating readiness.

Zhang et al. (2025); Jang et al. (2026)

04

## Multimodal & narrative

Figures, tables and equations enter the layout; story structure is now modelled explicitly.

Liang et al. (2025); Ozden et al. (2026a)

**Conclusion: generation is commoditised. Building one more generator is not a contribution.**

# Decomposing a PPT agent into 8 functions

A precise gap needs a function map — the next four slides walk through it

① **Parsing** → ② **Content** → ③ **Narrative** → ④ **Layout** → ⑤ **Output** → ⑥ **Inspection**

⑦ **Speaker notes** — Cross-stage: turn content into speakable pacing

⑧ **Provenance** — Cross-stage: every sentence needs a source

Next four slides: two gaps per slide — symptom · evidence · consequence

# Gap 1 | Parsing and content selection

① PARSING

## Figures and captions split apart

SYMPTOMS

- figure and caption are detached, so image and text no longer correspond
- two-column papers are read in the wrong order; paragraphs get spliced
- tables are flattened into a wall of text, losing row and column structure

EVIDENCE · APA

Parsing is treated as preprocessing and figure–text alignment is not enforced (Pang et al., 2025; Fu et al., 2021).

CONSEQUENCE

A corrupted input is inherited by every later step, and the error is invisible in the final deck.

② CONTENT SELECTION

## Even allocation, no emphasis

SYMPTOMS

- one slide per section, regardless of how much it matters
- the key experimental result is dropped while background survives
- method details are pushed into the introduction slide

EVIDENCE · APA

Most methods concentrate on text selection and ignore the link between space and importance (Liang et al., 2025; Chen et al., 2026).

CONSEQUENCE

The deck is complete but has no emphasis; the audience keeps only the title and the speaker cannot name the key claim.

# Gap 2 | Narrative and layout

③ NARRATIVE

## A table of contents, not a talk

SYMPTOMS

- section-by-section summarising reproduces the paper's order
- ten slides of background, two of results — emphasis inverted
- no motivation → problem → insight → evidence arc

EVIDENCE · APA

Source text is summarised straight into slides (Ozden et al., 2026a); explicit narrative planning remains rare (Vats et al., 2026).

CONSEQUENCE

The speaker reads a table of contents aloud and the audience cannot build causal understanding.

④ LAYOUT

## Overflow, overlap, clipping

SYMPTOMS

- text runs past the slide edge; elements overlap each other
- fonts too small and contrast too low to read once projected
- figures cropped, whitespace unbalanced, every slide a wall of bullets

EVIDENCE · APA

Local failures include overflow, overlap, clipping and off-canvas placement (Tian et al., 2026); models understand content yet fail at layout (Huang et al., 2025).

CONSEQUENCE

This is the most visible failure: text clipped and figures covered during the talk, so the audience loses trust immediately.

# Gap 3 | Output and inspection

⑤ OUTPUT

## Looks usable, cannot be edited

SYMPTOMS

- output is an image or HTML, not editable inside PowerPoint
- even .pptx output is all text boxes — editing one word breaks the layout
- charts are rasterised, so the underlying data is gone

EVIDENCE · APA

Agents frequently rasterise editable content or break the layout (Zheng et al., 2026b); only an object-model route preserves styling (Jung et al., 2025).

CONSEQUENCE

Users get an artefact they cannot modify, so it is unusable and the tool is abandoned.

⑥ INSPECTION

## One-shot feedback, no attribution

SYMPTOMS

- the whole deck is rewritten and then reviewed once
- errors cannot be localised to an element — which shape overflows?
- repair can make things worse, with no convergence guarantee

EVIDENCE · APA

A monolithic one-version-one-feedback loop makes local errors hard to attribute (Tian et al., 2026); reflection must be grounded in rendered state (Zheng et al., 2026a).

CONSEQUENCE

Iteration becomes a gamble: each pass may degrade the deck and nobody can say why.

# Gap 4 | Speaker notes and provenance

⑦ SPEAKER NOTES

## Notes are the bullets again

SYMPTOMS

- speaker notes duplicate the text already on the slide
- no time budget — no idea how many slides fit three minutes
- no transitions, so slides do not connect to each other

EVIDENCE · APA

Delivery is under-optimised in current systems (Yang et al., 2026); only 5 of 38 papers mention speaker notes, none as a research focus.

CONSEQUENCE

The speaker reads the slide while the audience reads and listens at once, splitting attention.

⑧ PROVENANCE

## Writes the words, cannot cite them

SYMPTOMS

- bullets carry no source, so there is no path back to a passage
- related work is presented as 'we propose'
- figures and tables misnumbered; numbers attached to the wrong claim

EVIDENCE · APA

Retrieval improves reliability but per-sentence provenance is not exposed (Xie et al., 2025); global intent is kept while sentence-level anchors are missing (Ozden et al., 2026a).

CONSEQUENCE

A wrong number goes unnoticed; in academic settings one misattributed claim destroys the credibility of the talk.

# A single failure is amplified downstream




Every upstream error is amplified downstream — and cannot be traced back

### Audience

Cannot follow or remember; sees clipped text live

→ Attention lost

### Speaker

Will not trust it; must rebuild; cannot answer 'where does this number come from?'

→ Trust collapses

### Field

Cannot compare systems, because there is no shared measure of trustworthiness

→ Progress stalls

**Measurement now exists (SlideAudit, DECKBench). What is missing is using it as a control signal — and measuring what it buys.**

# My direction: a paper-distillation PPT agent

Explicitly not lecture notes — the two tasks have different success criteria

| | Paper distillation (this project) | Lecture notes (not doing) |
|---|---|---|
| **Input** | A single paper (PDF) | Course material or lecture content |
| **Goal** | Compress the research into a defensible argument chain | Break knowledge into learning units |
| **Success** | Traceable content and a complete argument | Learner comprehension and exercises |
| **Hard part** | Factual correctness | Instructional design |

**RESEARCH QUESTIONS**

**RQ1** How can every bullet be anchored to a verifiable source?

**RQ2** How can deck usability be turned into a computable metric?

**RQ3** Can an inspect–repair loop improve both at an acceptable cost?

# Three contributions

## 01 A computable notion of trustworthiness

Content trust and deck usability become measurable, and the measurements are used as control signals inside generation rather than as scores applied after it.

**Deliverable:**
**metric definitions + open-source toolkit**

## 02 A step-wise inspect–repair loop

Change one element at a time, render and verify immediately, so errors are attributable; report convergence rather than only a final score.

**Deliverable:**
**agent implementation + convergence curves**

## 03 A public dataset and evaluation protocol

150 human-annotated reference slides, a 30-participant pairwise study, and rerunnable evaluation scripts.

**Deliverable:**
**dataset + benchmark script**

# Operational definition: deck-ready rate

DRR = fraction of slides passing every gate; each threshold is standard-anchored or empirically calibrated

| Gate | Rule | Threshold | Anchor | Swept over |
|---|---|---|---|---|
| **G1 Containment** | Every shape box inside the slide; no image cropped | 0 pt | binary rule | — |
| **G2 Overlap** | Text–text intersection ÷ area of smaller box | > 5% | empirical | 2 / 5 / 10% |
| **G3 Font size** | Body text and titles, read from run properties | 18 / 28 pt | design guidance | 16 / 18 / 20 |
| **G4 Contrast** | WCAG 2.2 AA ratio, from fill and slide background | 4.5:1 (3:1 large) | published standard | — |
| **G5 Text fit** | Rendered text height ÷ box height, real font metrics | ≤ 0.95 | renderer tolerance | 0.90 / 1.00 |
| **G6 Density** | Characters per slide; bullets per text block | ≤ 140; ≤ 6 | empirical | 120 / 160 |

Calibration: sweep each threshold, keep the value that maximises agreement with human labels, report the Spearman rho curve.

Positioning: SlideAudit supplies the label space (image-based critique); DECKBench scores decks against references; DRR is reference-free and object-level — a control signal, not a score.

# Evidence plan: agreement, power, fallbacks

The grade depends on whether a sceptic can rerun these three numbers

### DETECTOR VALIDATION

- 150 slides, stratified across all three systems — not only our own output
- 2 independent annotators; rulebook training plus 10 calibration slides
- Cohen's kappa >= 0.70 per class, disagreements adjudicated by a third pass
- Report precision / recall / F1 per class, never an aggregate only
- Frozen before the human study; if per-class F1 < 0.70 on calibration, revise the rule and re-freeze

### HUMAN STUDY

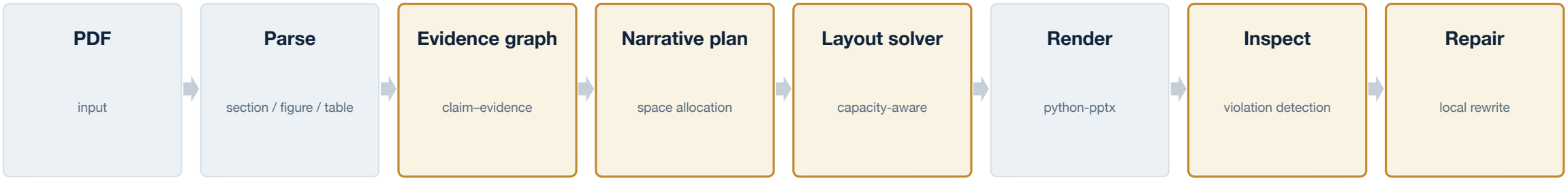
- Within-subject: 30 raters x 4 papers, all three systems on the same paper
- One primary contrast pre-registered: ours vs the VLM-judge loop
- Holm–Bonferroni across the remaining two comparisons
- n = 30 detects d about 0.55–0.60 at 80% power; report CI and effect size
- Raters blind to condition order; papers rotated to control learning effects

### IF RECRUITMENT FALLS SHORT OF 30

3 expert raters with ICC(2,k) · expand automated QA from 10 to 25 items per deck · report the posterior probability of superiority · label the result exploratory, not confirmatory

# System architecture

Gold = new contributions; the rest reuses validated practice




### Restrict the output space

The model writes structured content only; geometry and styling belong to templates

### One element at a time

Local rewrites with immediate verification make errors attributable

### Every judgement has a number

No 'looks fine' — only measured violations and ratios

The output is not an image but an editable .pptx: layout comes from constraint solving, content from the evidence graph.

# How I will show it works

Three tracks in parallel, avoiding the circularity of an LLM judging an LLM

| Track | Metric | Sample | Why it is needed |
|---|---|---|---|
| **Automatic (geometry)** | Violation count, deck-ready rate | 8 held-out papers, 150 annotated slides | Deterministic and rerunnable |
| **Task-based (QA)** | Answer accuracy from the deck | 10 questions × 8 papers | Tests whether a reader can extract correct facts |
| **Human** | Pairwise preference, speakability | 30 participants | Design has no ground truth, so only relative comparison is valid |

**BASELINES — EACH ANSWERS A DIFFERENT QUESTION**

1 **Naive single-pass** — does inspection add anything at all?

2 **PPTAgent** — versus the strongest reproducible generator

3 **VLM-judge repair loop** — computable rules versus subjective judgement

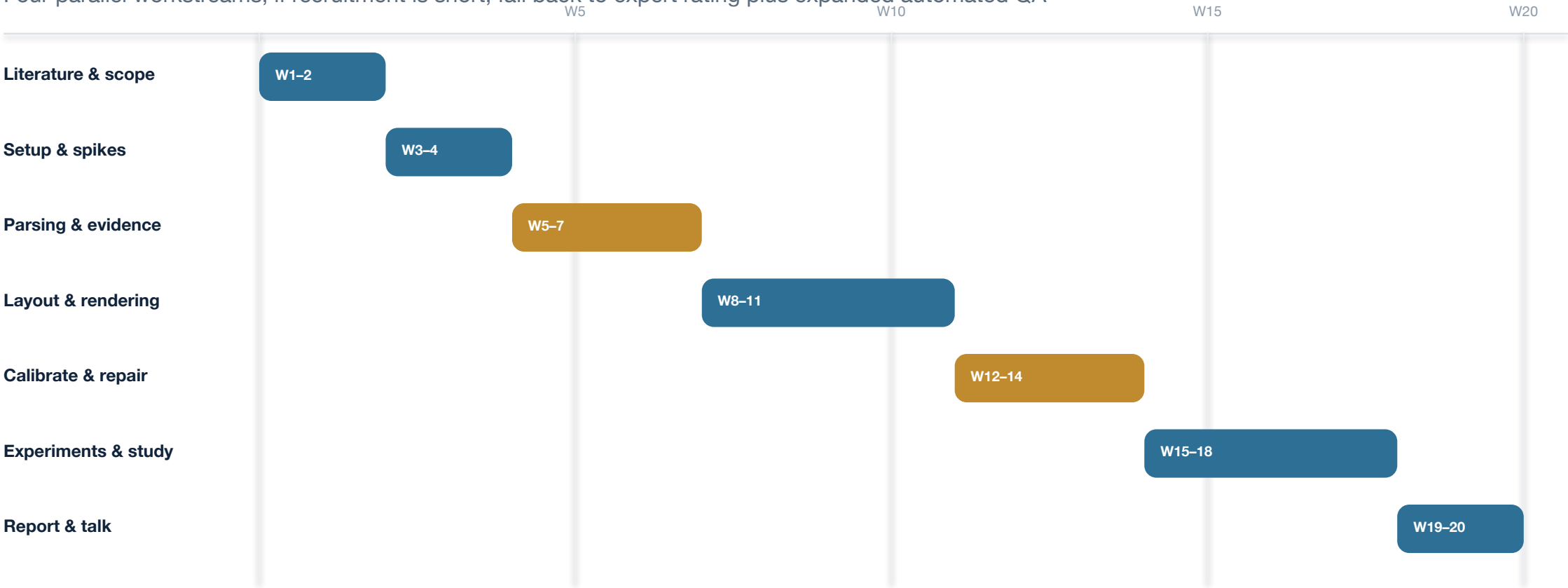
**3 is the primary claim: the same loop, verification by computable rules instead of a VLM judge.**

**ABLATION**

Remove anchors
Remove inspection
Remove repair
→ per-module attribution

# 20-week timeline

Four parallel workstreams; if recruitment is short, fall back to expert rating plus expanded automated QA




**2026-11-05**

Proposal submission

**2027-01-07**

Interim Report

**2027-04-08**

Final Report + Turnitin + AI Report

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